21. Textiles and leather — fibres, spinning, weaving, dyeing, sewing, tanning and shoemaking

Oracle OS guide. Manuals in the library: "Hand-Loom Weaving", "Textiles" (Dooley), "Textiles and Clothing", "Vegetable Dyes", "A Text-book of Tanning", "Leather" (Adcock); Gutenberg technology (illustrated).

Fibres

- **Wool** (sheep, goat, angora rabbit): shear in spring, wash in warm soapy water without rubbing (it felts), dry.
- **Flax**: sow in spring, pull when yellow, **ret** (rot the stems in water or dew 2–4 weeks), dry, break and scutch to free the fibre, hackle (comb).
- **Nettle and hemp**: like flax; strong fibre for rope and sacks.
- **Reuse**: unravel old fabric and knitwear — the yarn already exists.

Carding and spinning

- Hand cards (nail brushes) align fibres into rolls.
- **Drop spindle** (stick with a weight): attach fibre, spin the spindle and let twist climb while drawing the fibre out slowly; wind on and repeat. A spinning wheel is faster.
- **Plying** two yarns in the opposite direction gives stronger, balanced yarn.

Weaving, knitting and sewing

- **Simple loom** (frame or rigid heddle): stretch the warp (vertical threads) and pass the weft over and under, beating it tight. See "Hand-Loom Weaving".
- **Knitting and crochet** with wooden or bone needles; socks, hats, jumpers.
- **Sewing**: backstitch (strong), running stitch, patches; needles from bone or sharpened, drilled wire.
- **Felt**: wool + hot water + soap + friction → dense fabric for boots, hats and blankets.

Dyeing

Mordant first (alum, or an iron pot), then a hot dye bath for 1 h: onion skins (yellow/orange), walnut hulls and leaves (brown), madder (red), woad/indigo (blue), heather and broom (yellow), iron (darkens everything).

Tanning hides

1. **Skin** carefully; **flesh** (scrape fat and meat) over a log.
2. **Hair off** (leather): soak in water with ash or lime 3–7 days until hair slips; scrape.
3. **Bark tanning** (tannin): ground oak, chestnut or willow bark in water; hide in it 4 weeks to several months (thick leather), strengthening the solution gradually. Gives hard, lasting leather (shoes, belts, harness).
4. **Brain/oil tanning** (soft hide, buckskin): spread a paste of cooked brain or egg yolk, work the hide by stretching as it dries, **smoke** it so it doesn't stiffen in rain.

5 Dress with tallow or oil.

Shoes

Thick leather sole (or old tyre), soft leather upper, sewn with waxed linen thread or leather thongs; pattern made from the foot. Tyre-and-strap sandals are quickest. Dry socks and footwear are essential for walking and working (see "Stretching and pain" — feet).